

# ControlPlane.ai — Admission-Control Layer for AI That Acts

Accenture Innovation Challenge 2026 · Round 2 · Team ControlPlane · PS #1

## 1. What this is

Every AI response that asks permission to **act** — issue a refund, send an email, publish a record — is a set of **claims**. ControlPlane captures provenance **outside** the model (STEP → SPAN → CLAIM → ACTION) and admits or holds each action through a frozen interlock matrix. Unproven claims cannot authorize irreversible actions. Lane 1 is deterministic; no LLM sits on the critical path.

This repository is the working prototype. The numbers in the evidence section are produced by `make eval` and `make bench` and are regenerated on every run — they are measured, not asserted.

## 2. The mechanism

Run any response text through `extract` → `bind` → `entitle` → `interlock` → `shadow`:

- **extract** — Lane-1 segmenter produces typed claims (numeric / structural / temporal / textual / derived) with hedging and role-in-action, no LLM.
- **bind** — each claim is matched to provenance spans: BM25 + lexical gate (textual), Indian/Western numeric normalization (numeric), symbol-table lookup (structural), recompute (derived). Middle band → UNKNOWN, never a default pass.
- **entitle** — span ACL is checked against principal clearance; an unbound PII entity is a leak (Rule A) → Block at R2/R3.
- **interlock** — a frozen 16-cell matrix maps (blast-tier × verdict-column) to one of five actuators. The matrix is never redrawn.
- **shadow** — every decision is dual-emitted; the published FNR carries a Wilson interval.

## 3. Run it in 60 seconds (clean clone)

```
git clone <repo> controlplane-ai && cd controlplane-ai
python3 -m venv .venv && source .venv/bin/activate
pip install -e "[dev]"
bash scripts/preflight-lite.sh          # must PASS
pytest -q                             # 330+ green
make eval                              # real FNR with Wilson CI
make bench                             # n=10000 per-stage latency
uvicorn controlplane.server.app:create_app --factory --port 8787
# open http://127.0.0.1:8787/gate and paste your own response text
```

The `/gate` console runs the real pipeline on arbitrary text — no fixtures, no canned scenario.

## 4. What to look at

- **/gate** — paste a response; watch claims, binding method + rationale, symbol table, matrix cell, actuator, evidence packet, dead compute, per-stage latency.
- **examples/refund\_trace\_demo.py** — Edit + Escalate (held) reproduced with fixtures off.
- **evals/cases/** — 150+ labelled cases incl. hard negatives; this is the corpus a judge can attack.
- **controlplane/interlock.py** MATRIX — the sixteen transcribed cells, sole decider.

## 5. Capability ledger

Every mechanism in the architecture, marked honestly. This preempts 'is any of this real?' by answering first — the same move as publishing the FNR.

| Mechanism                              | Status                       | Where                       |
|----------------------------------------|------------------------------|-----------------------------|
| Claim extraction (Lane 1)              | implemented and tested       | controlplane/extract.py     |
| Numeric normalization (IN/US grouping) | implemented and tested       | controlplane/numeric.py     |
| BM25 + lexical entailment gate         | implemented and tested       | controlplane/bm25.py        |
| Symbol table / structural lookup       | implemented and tested       | controlplane/symbols.py     |
| Binder v2 route dispatch               | implemented and tested       | controlplane/binder.py      |
| PII Rule A (leak → Block)              | implemented and tested       | controlplane/pii.py         |
| Entitlement / ACL audit                | implemented and tested       | controlplane/entitlement.py |
| Frozen 16-cell interlock matrix        | implemented and tested       | controlplane/interlock.py   |
| Lane deadlines (async gather)          | implemented and tested       | controlplane/lanes.py       |
| Override → chained ledger → labels     | implemented and tested       | controlplane/feedback.py    |
| Eval harness + Wilson CIs              | implemented and tested       | evals/run.py                |
| Load bench n=10000                     | implemented and tested       | scripts/load_bench.py       |
| Shadow-replay threshold gate           | implemented and tested       | controlplane/feedback.py    |
| Canary auto-rollback                   | implemented and tested       | controlplane/feedback.py    |
| Jurisdiction packs (EU/IN/GDPR)        | implemented and tested       | policies/*.yaml             |
| Bias counterfactual (Lane 3)           | implemented, prototype-grade | controlplane/bias.py        |
| Multi-turn claim inheritance           | implemented and tested       | controlplane/session.py     |
| GPU / trained NLI adapter              | designed, not built          | Lane-2 slot (stub)          |

## 6. Evidence (measured on this run)

Eval corpus: **168** labelled cases (426 bindings). Ungrounded catch rate (FNR) **1.1%** (95% CI 0.2%–5.8%) → we HOLD **98.9%** of ungrounded responses. Passable-action false-hold (FPR) **0.0%** (95% CI 0.0%–15.5%). Fail-closed caution on hard-negatives **64.2%**. Abstention (UNKNOWN rate) **0.5%** of bindings. Irreversible actions (R2/R3) escalated by design: held=0, passed=0.

Per-route (Wilson 95% CI):

| Route      | n   | supported | unknown | contradicted | unsupported | abstain |
|------------|-----|-----------|---------|--------------|-------------|---------|
| numeric    | 119 | 105       | 0       | 14           | 0           | 0.0%    |
| structural | 110 | 178       | 0       | 0            | 33          | 0.0%    |
| textual    | 71  | 50        | 2       | 0            | 19          | 2.8%    |
| derived    | 12  | 2         | 0       | 10           | 0           | 0.0%    |
| temporal   | 11  | 11        | 0       | 0            | 0           | 0.0%    |
| none       | 2   | 0         | 0       | 2            | 0           | 0.0%    |

Load bench: n=10000 at concurrency sweep [1, 4, 16, 64]. Per-request compute p50=0.435 ms, p95=0.5173 ms, p99=0.6174 ms, mean=0.4493 ms, max=8.0535 ms. Best throughput 2192.3 rps. (In-process compute, not API latency; 40ms is a Lane-1 target, never a measured p95.)

Refund language in this system is **held / escalated with an evidence packet** — never 'blocked'. Irreversible actions are not released without provenance.

## 7. Repo map

controlplane/ — pipeline (extract, binder, symbols, bm25, numeric, pii, entitlement, interlock, lanes, feedback, bias, shadow, server)

evals/ — corpus (cases/), harness (run.py), Wilson CIs

policies/ — jurisdiction packs (eu / in / gdpr / default)

scripts/ — bench, proposal/readme PDF builders, preflight

examples/ — refund\_trace\_demo, knowledge\_flip\_demo, multi\_usecase\_demo

submission/ — proposal PDF, pitch deck, latency bench, SBOM

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